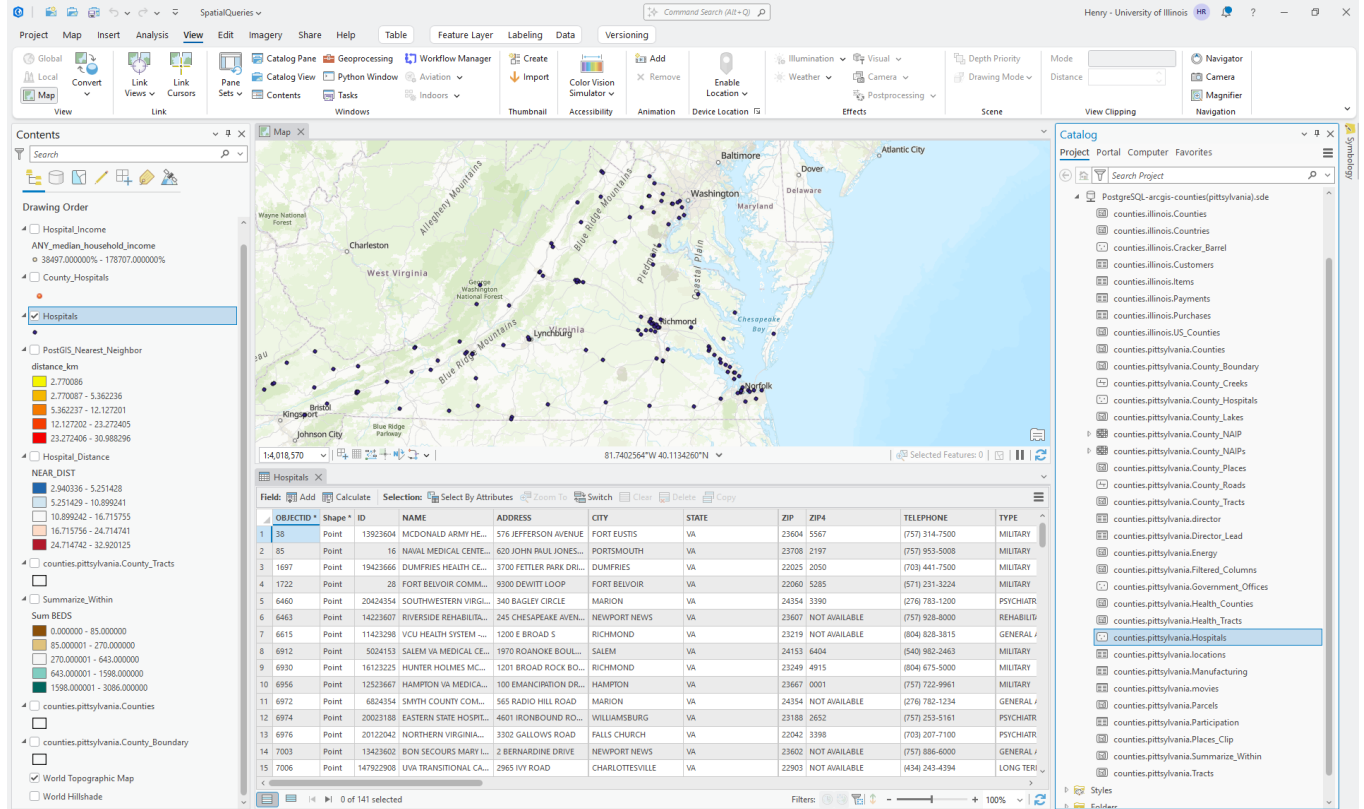


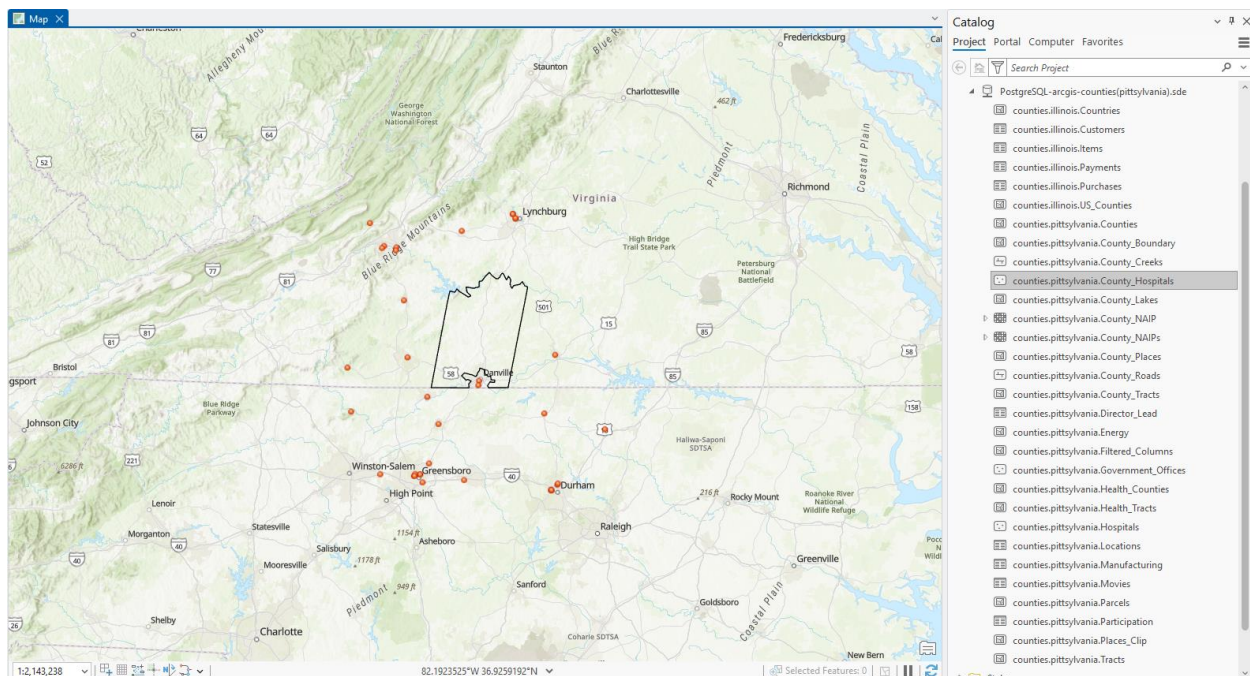
Week 10 – Spatial Queries Revision

Henry Rotunno

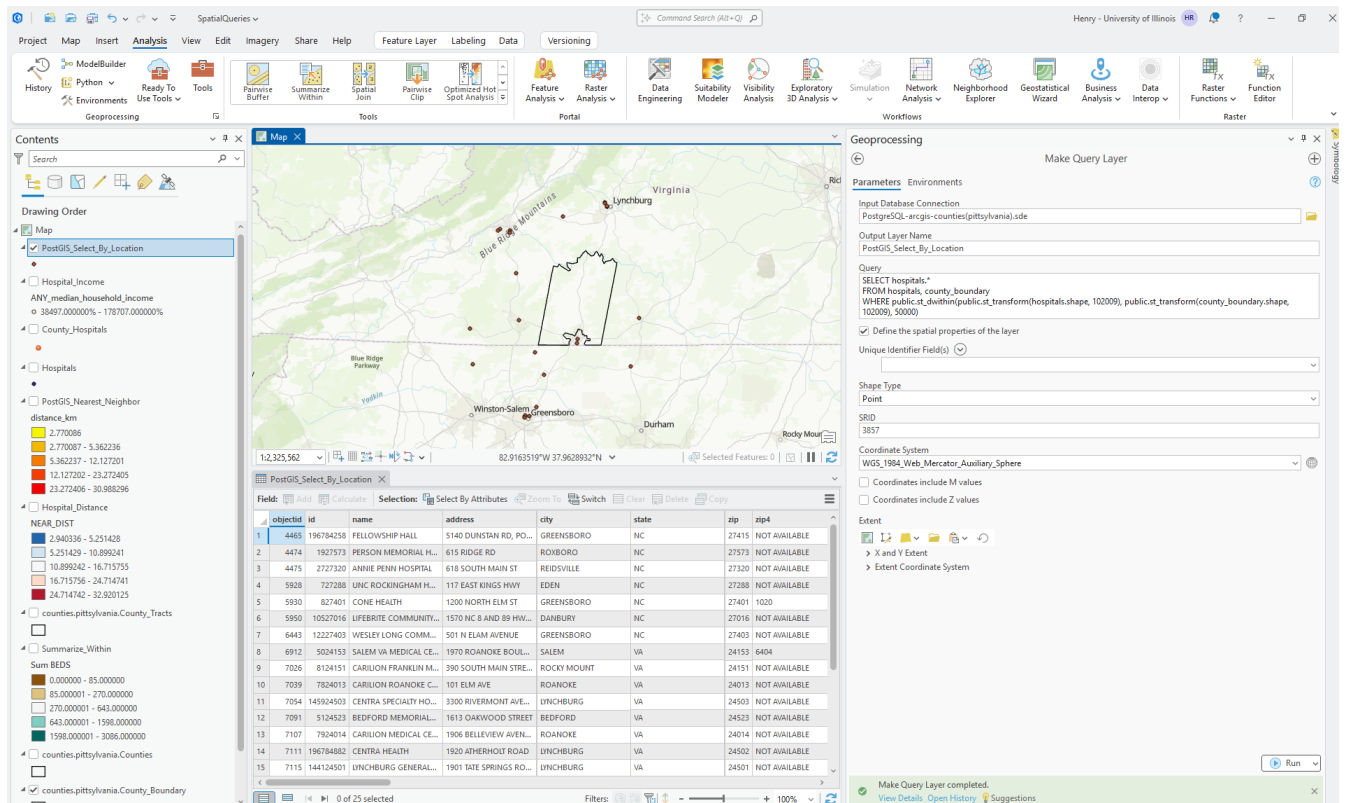
31 October 2025



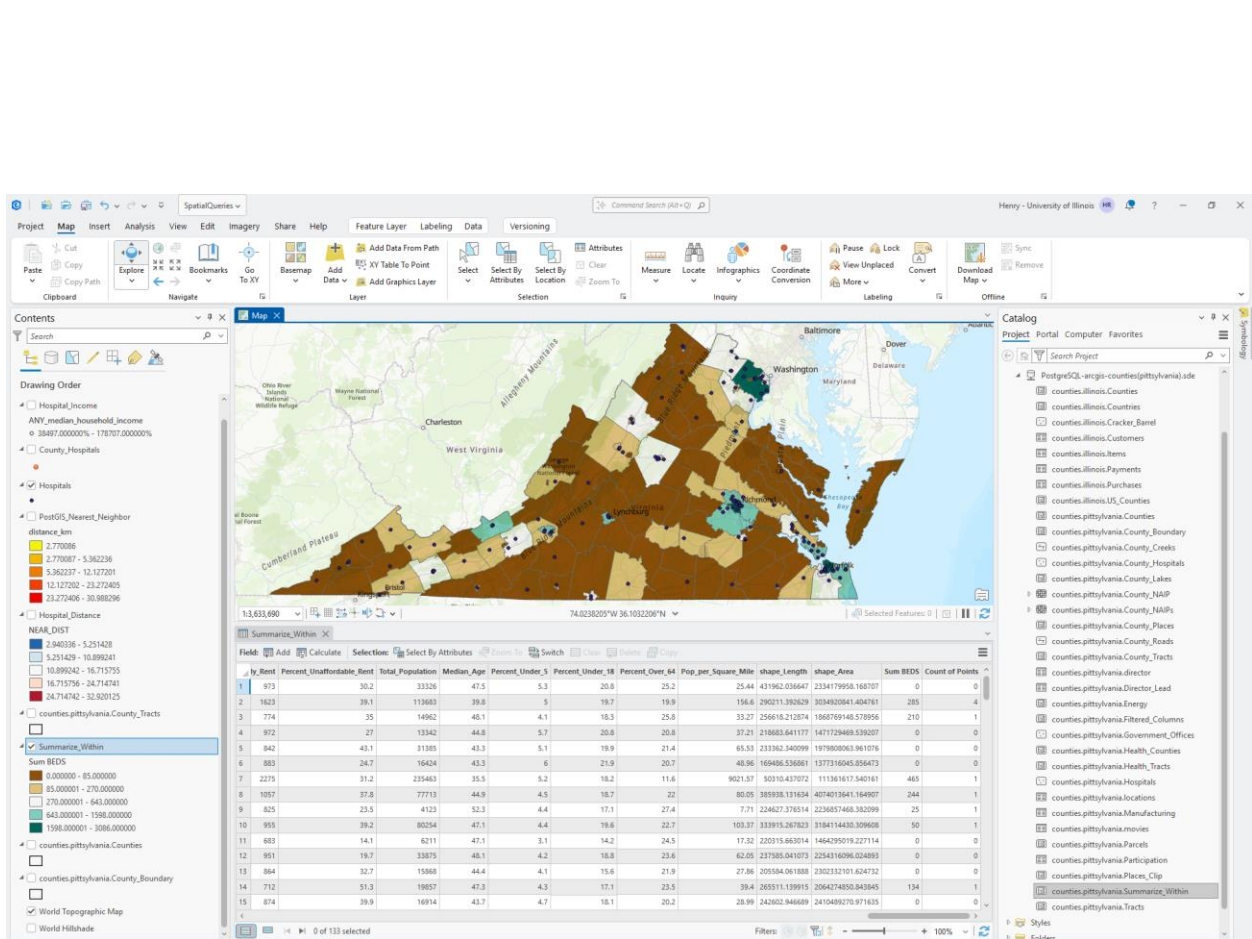
Hospitals (US Department of Homeland Security n.d.)



Select by location



PostGIS select by location



Summarize within

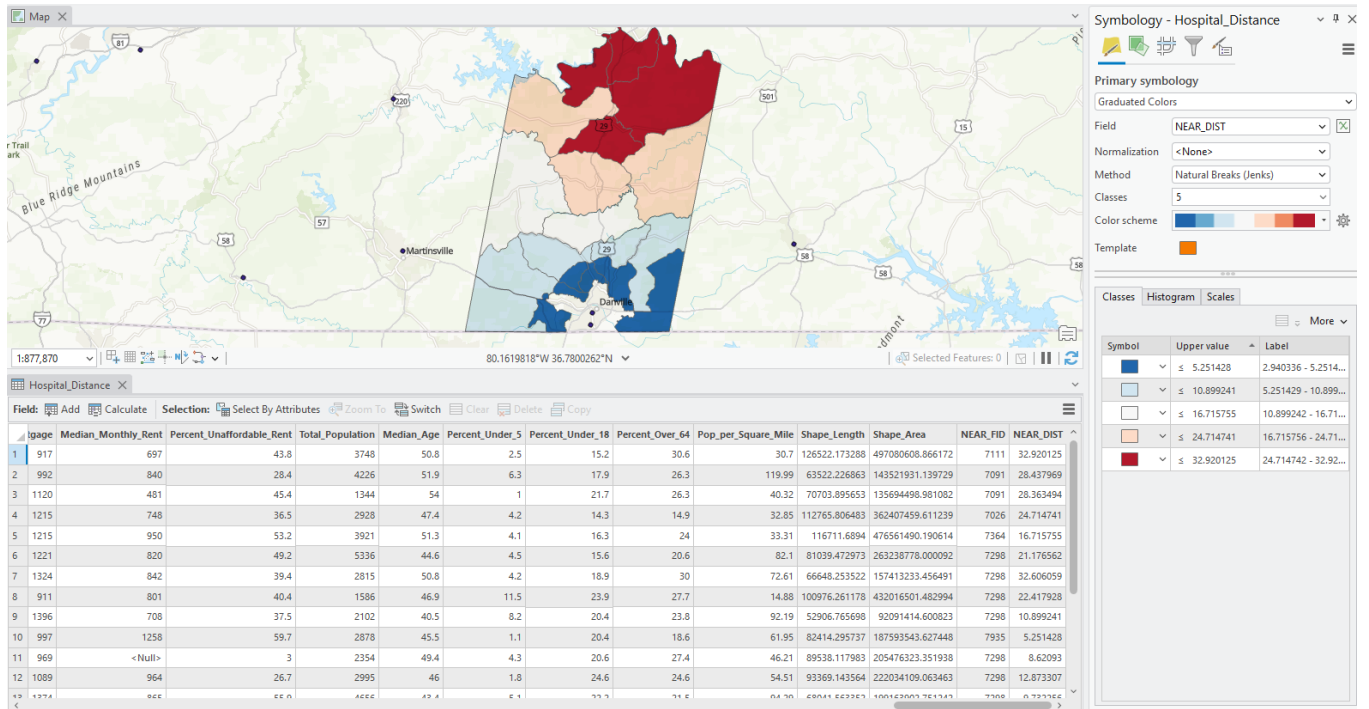
```

SELECT counties.name, SUM(state_hospitals.beds)
FROM counties
JOIN state_hospitals
ON public.st_intersects
(public.st_transform(counties.shape, 4326),
public.st_transform(state_hospitals.shape, 4326))
GROUP BY counties.name

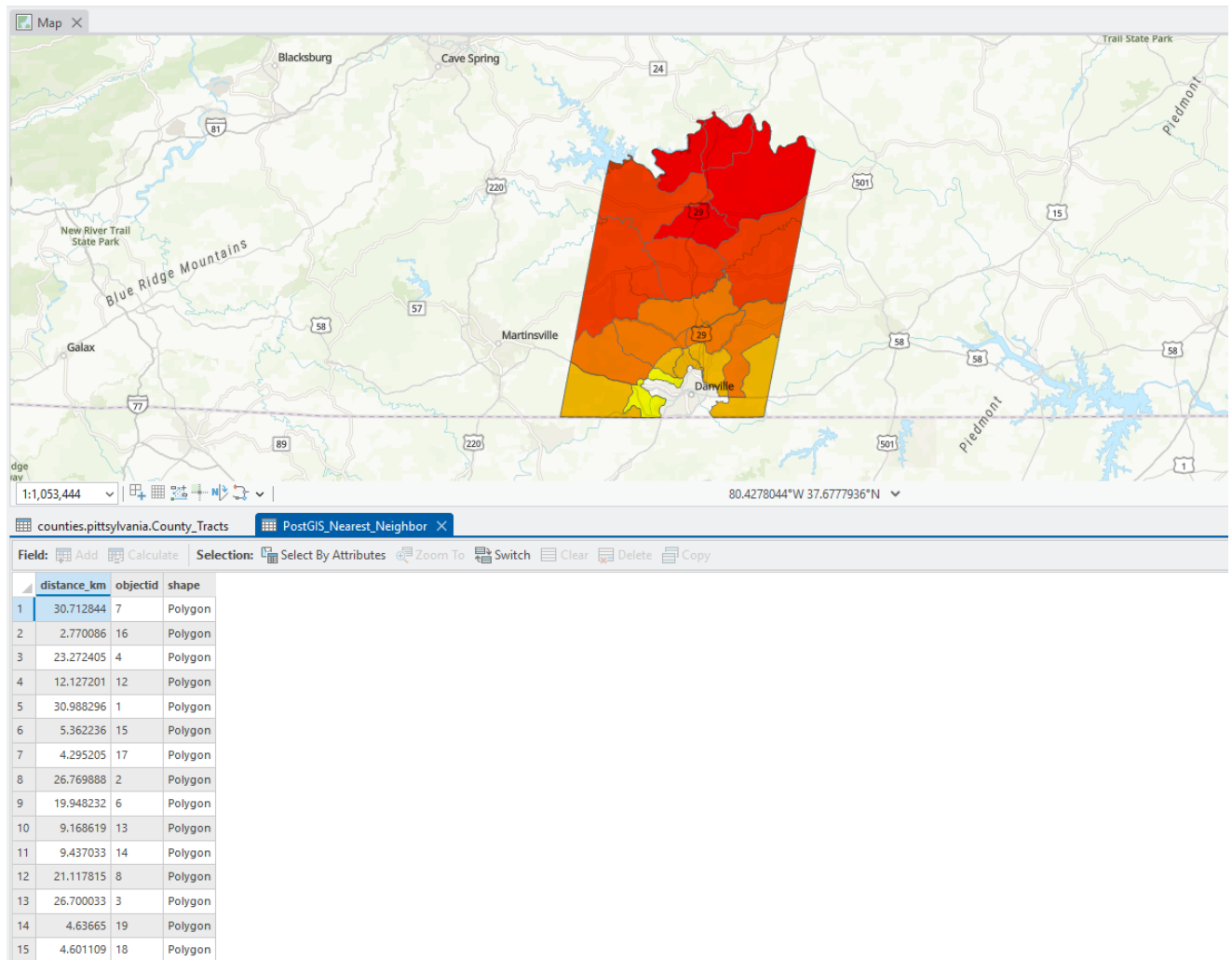
```

name	sum
Albemarle	285
Alexandria	312
Alleghany	210
Arlington	465
Augusta	244
Bath	25
Bedford	50
Bristol	53
Buchanan	134
Charlottesville	643
Chesapeake	310
Chesterfield	841
Culpeper	70
Danville	322
Dickenson	25
Dimwiddle	126
Emporia	80
Essex	67
Fairfax	2224
Fauquier	109
Franklin	142
Fredericksburg	1039
Galax	141
Giles	728
Gloucester	87
Gooschland	114
Halifax	192
Hampton	636
Hanover	265
Harrisonburg	270
Henrico	1598
Hopeveall	178
James City	366
Lancaster	35
Lee	70
Lexington	25
Loudoun	615
Lynchburg	1489
Manassas	150
Martinsville	220
Mecklenburg	70
Montgomery	330
New Kent	94
Newport News	536

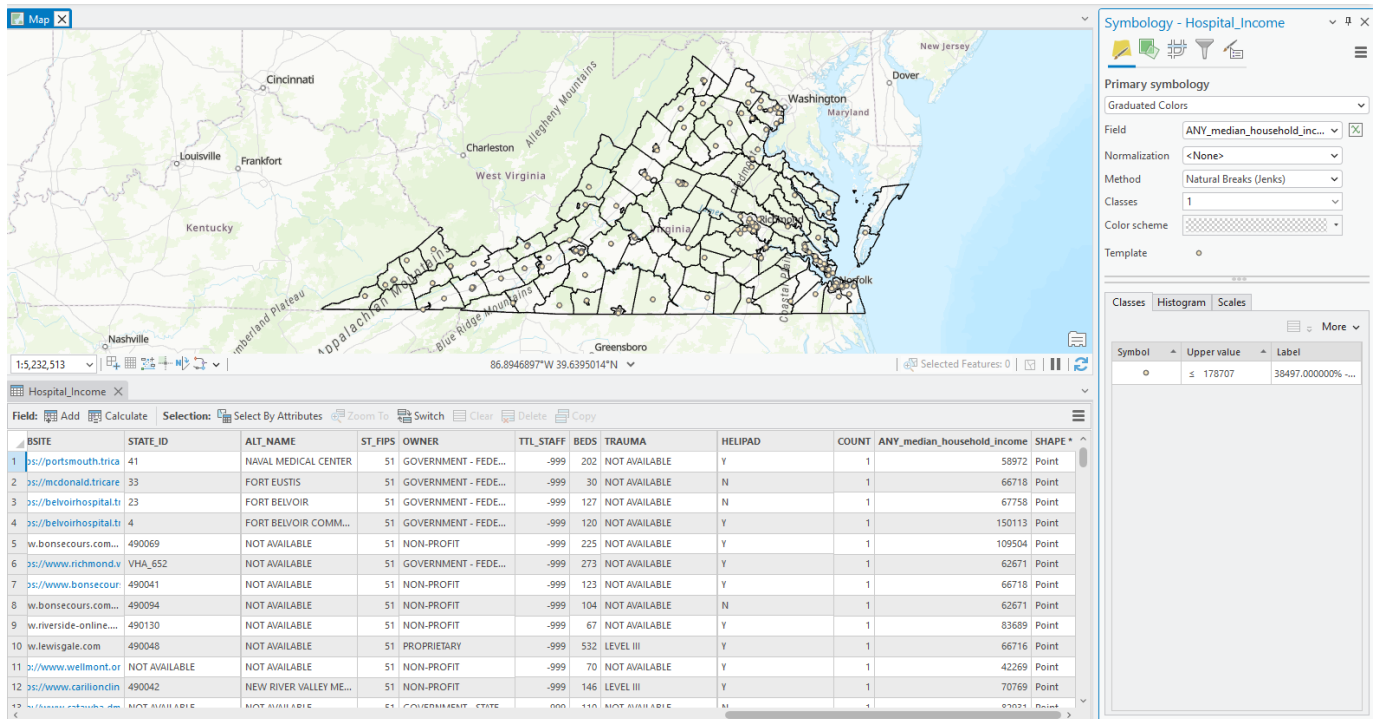
PostGIS summarize within using ST_Intersects by GROUP BY



Near



PostGIS nearest neighbor



Spatial join

```
SELECT counties.Median_Household_Income, County_Hospitals.*
FROM County_Hospitals
INNER JOIN counties ON
public.st_within(
public.st_transform(County_Hospitals.shape, 182809),
public.st_transform(counties.shape, 182809))
```

median_household_income	objectid	id	name	address	city	state
77644.00000000	20	5124523	BEDFORD MEMORIAL HOSPITAL	1613 OAKWOOD STREET	BEDFORD	VA
68348.00000000	17	8124151	CARILION FRANKLIN MEMORIAL HOSPITAL	390 SOUTH MAIN STREET	ROCKY MOUNT	VA
49244.00000000	25	11824592	SENTARA HALIFAX REGIONAL HOSPITAL	2204 WILBORN AVENUE	SOUTH BOSTON	VA
50938.00000000	31	8324171	PIONEER COMMUNITY HOSPITAL	18688 JEB STUART HIGHWAY	STUART	VA
82931.00000000	26	4724070	CATAWBA HOSPITAL	5525 CATAWBA HOSPITAL DRIVE	CATAWBA	VA
42778.00000000	28	7624541	SOVAH HEALTH - DANVILLE	142 SOUTH MAIN STREET	DANVILLE	VA
42778.00000000	35	20324541	SOUTHERN VIRGINIA MENTAL HEALTH INSTITUTE	382 TAYLOR DRIVE	DANVILLE	VA
59808.00000000	19	145924503	CENTRA SPECIALTY HOSPITAL	3300 RIVERMONT AVENUE	LYNCHBURG	VA
59808.00000000	22	196784882	CENTRA HEALTH	1920 ATHERHOLT ROAD	LYNCHBURG	VA
59808.00000000	23	144124501	LYNCHBURG GENERAL HOSPITAL	1901 TATE SPRINGS ROAD	LYNCHBURG	VA
59808.00000000	24	5624503	VIRGINIA BAPTIST HOSPITAL	3300 RIVERMONT AVENUE	LYNCHBURG	VA
42434.00000000	30	8224112	SOVAH HEALTH - MARTINSVILLE	320 HOSPITAL DRIVE	MARTINSVILLE	VA
52671.00000000	18	7824013	CARILION ROANOKE COMMUNITY HOSPITAL	101 ELM AVE	ROANOKE	VA
52671.00000000	21	7924014	CARILION MEDICAL CENTER	1906 BELLEVIEW AVENUE	ROANOKE	VA
66716.00000000	15	5024153	SALEM VA MEDICAL CENTER	1970 ROANOKE BOULEVARD	SALEM	VA
66716.00000000	29	4824153	LEWISGALE MEDICAL CENTER	1900 ELECTRIC ROAD	SALEM	VA

16 rows (1.583 s) Edit, Explain, Export

PostGIS spatial join